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HEAS: A HIERARCHICAL EVOLUTIONARY AGENT-BASED SIMULATION FRAMEWORK FOR MULTI-OBJECTIVE POLICY SEARCH

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ABSTRACT

Agent-based model optimization pipelines often recompute the same outcome metric across optimization, tournament evaluation, and statistical validation using independently implemented aggregation logic at each pipeline stage. This silent aggregation divergence is a structural risk in loosely-coupled simulation pipelines: policy rankings can differ across stages without any explicit error. We propose HEAS, a composable framework that eliminates this risk through metric contracts—a runtime-enforceable design pattern that makes divergence provably impossible on all contract-verified execution paths. HEAS combines a uniform `metrics_episode()` contract with a Layer/Stream/Arena composition model, a pluggable optimizer interface, and a tournament primitive for multi-scenario ranking. Three case studies—ecological, enterprise, and a Wolf-Sheep ecological scenario—validate composability across the targeted model family. Against a Mesa 3.3.1 baseline, HEAS reduces coupling code by 97% (160 to 5 lines). In the tested ecological scenarios, the evolved champion wins all 32 held-out and all 24 single-anchor OOD cases.

1 INTRODUCTION

Agent-based models are foundational for policy design across ecology, economics, and organizational science, yet coupling an ODE-style or tabular ABM to evolutionary optimization requires integration infrastructure that general-purpose frameworks do not provide. This paper addresses a specific integration failure: when researchers link an ABM to an evolutionary algorithm, the same outcome metric is recomputed in incompatible ways across three separate code paths—optimization, tournament evaluation, and statistical validation—introducing silent divergence that is difficult to audit. Current practice resolves this through bespoke coupling code (roughly 160 lines in a representative Mesa workflow) that entangles metric definition with solver, comparison, and inference logic.

We introduce HEAS, a framework that standardizes metric definition through a uniform `metrics_episode()` contract consumed identically by the evolutionary optimizer, tournament scorer, and bootstrap confidence-interval pipeline. This approach reduces coupling code to five lines and eliminates conditions for metric divergence.

HEAS targets a specific niche—coupled ODE/tabular dynamics, multi-objective evolutionary search, and multi-run validation—rather than the full ABM space including spatially-explicit heterogeneous-agent systems. The framework's four core components (C1–C4, Section 2) are independently useful and may be adopted in isolation.

1.1 The Coupling Code Problem

Consider a researcher using Mesa to study regulatory policy in a multi-firm economy. She wants to (1) evolve a Pareto-optimal tax regime using NSGA-II, (2) compare the evolved regime against a reference policy across independent economic scenarios using tournament voting, and (3) report bootstrap confidence intervals. Each step reads the same welfare metric, yet in Mesa each is an independent code path: the DEAP fitness function extracts welfare from the model's `DataCollector` after each run; the tournament scorer extracts welfare from a separate episode-evaluation function, possibly with different aggregation; and the bootstrap CI loop extracts welfare from a third data collection loop, potentially inconsistent with both.

In practice, silent divergence is difficult to detect. Consider an experiment where the tournament scorer reads the final step's biomass while the evolutionary algorithm reads the episode mean biomass. The evolved champion may become the policy with volatile-but-high peaks rather than sustainable mean production—a consistently applied but undetected inconsistency that could persist through publication. Such divergences are common precisely because *no single code path* connects metric definition to all three consumers.

When the researcher adds a second objective—welfare inequality—she must edit the DataCollector reporter, the DEAP fitness function, the tournament scorer, and the CI loop, keeping all four in sync manually. A silent divergence can persist undetected for weeks and invalidate a publication. This pattern, which we call *coupling code*, appears in representative ABM-based policy-search workflows (Pangallo, Heinrich, and Farmer 2019; Capri, Ignaccolo, Inturri, Pluchino, and Rapisarda 2023).

1.2 Contributions

This paper identifies silent aggregation divergence as a structural risk in loosely-coupled ABM pipelines and proposes metric contracts as a composable design pattern for eliminating it. The pattern enforces a uniform `metrics_episode()` interface consumed identically across all pipeline stages, making divergence provably impossible on all contract-verified execution paths—by construction, not post-hoc validation.

HEAS enforces structural consistency: the same dispatch-time metric contract is verified across all pipeline stages. It does not enforce semantic validity: an incorrect but consistently applied `metrics_episode()` will pass contract verification. Semantic correctness—whether the metric computation is logically sound for the scientific question—remains the modeller's responsibility.

To realize this pattern, HEAS provides four implementation components: **C1** defines the contract (`metrics_episode()` returning `dict[str, float]`); **C2** (Layer/Stream/Arena composition) enables models to produce C1 without rewriting dynamics; **C3** (pluggable multi-objective optimizer wrapper) and **C4** (tournament with four voting rules) consume C1 without internal variation. These components are independently replaceable, but none are separate contributions—they are infrastructure for enforcing the contract.

The methodological contribution is twofold: (a) a replicable framework design that reduces per-project coupling code from 160 LOC to 5 LOC (97% reduction against Mesa) and eliminates metric divergence risk through single-source-of-truth metric enforcement, and (b) a validated reference implementation with empirical evidence (controlled aggregation experiment with stochastic robustness test, §5.2; cross-domain tournament validation, §5.5; domain-internal robustness tests, §5.5) that the contract mechanism prevents rank reversals and supports multi-objective ABM workflows at scale. This combination of architectural design and validated effectiveness provides both a learning artifact and an extensible platform for simulation-based policy search.

2 RELATED WORK

Agent-based modeling is widely used in ecology and economics, yet frameworks like Mesa do not natively support coupled evolutionary optimization and statistical comparison (Kazil, Masad, and Crooks 2020). Simulation optimization recognizes this integration gap (Fu 2002), but search libraries (DEAP, Optuna) and ranking methods do not standardize metric sharing across optimization, tournament, and inference stages (Fortin, De Rainville, Gardner, Parizeau, and Gagné 2012; Akiba, Sano, Yanase, Ohta, and Koyama 2019).

Of the incumbent ABM and optimization frameworks, none enforces a unified metric contract at the framework level: EMAworkbench provides ensemble orchestration but leaves metric-stage consistency to per-project code; Optuna and Mesa 3.3 make no provision for shared metric definitions across optimizer,

Table 1: HEAS contract architecture. `metrics_episode()` is the single aggregation callable shared by all pipeline stages; without it, each stage independently selects its aggregation field, producing silent rank reversals.

Component	**Stage**	**Role / contract interface**
Layer	Simulation	Atomic unit; exposes `metrics_episode()` → `dict[str, float]`
Stream	Simulation	Named group of Layers; namespaces metric keys
Arena	Simulation	Orchestrates Streams; dispatches unified contract
Optimizer	Pipeline	NSGA-II EA; reads `metrics_episode()` objective keys
Tournament	Pipeline	Cross-scenario ranking (argmax/Borda/-Copeland/majority)
Inference	Pipeline	BCa CI; reads same keys — no re-aggregation

tournament, and inference stages. HEAS is the only framework that enforces `metrics_episode()` as a dispatch-time type-checked contract across all three stages.

HEAS is positioned at this interface. It does not propose a new optimizer. Instead, it standardizes the metric pathway shared by optimization, tournament scoring, and inference—narrower in scope and more specific in its methodological target than general-purpose ensemble frameworks.

Contract-as-methodology tradition. The ODD protocol (Grimm, Berger, Bastiansen, Eliassen, Ginot, Giske, Goss-Custard, Grand, et al. 2006) established author-declared documentation contracts for ABM, but compliance cannot be machine-verified at execution time, leaving aggregation choices invisible to the contract. Thiele & Grimm (Thiele and Grimm 2015) formalized replication as community enforcement; failure is detected *post-hoc* only. Grimm & Railsback's POM (Grimm and Railsback 2012) extended contracts to outcome specification yet remained post-hoc and researcher-driven: it constrains *what* outcomes matter without constraining *how* they are computed across pipeline stages. Each framework remained prescriptive and external to the execution engine. HEAS closes this gap by encoding the metric contract as a type-checked callable signature verified at dispatch time—divergence becomes provably impossible on all contract-verified execution paths, not by documentation obligation but by construction.

3 THE HEAS FRAMEWORK

HEAS is implemented in Python 3.11 and depends on DEAP for evolutionary algorithms, NumPy for numerical computation, and SciPy for statistical utilities. All components—EA algorithm, voting rule, statistical estimator—can be replaced without changing simulation or metric contract code. Table 1 summarises the full stack.

3.1 Layered Composition

This subsection defines the compositional boundary that lets models be reused without rewriting evaluation code. The fundamental abstraction is a `Layer` with four methods (`step`, `metrics_step`, `metrics_episode`, `reset`). All downstream components read the same `metrics_episode()` keys.

Layers compose into **Streams**—named groups forming a coherent sub-simulation—and Streams compose into an **Arena**, which orchestrates step ordering, inter-stream signal routing, and episode-level aggregation. The naming convention `stream_name.metric_name` ensures globally unambiguous keys regardless of Arena size. Explicit composition boundaries make components testable and replaceable without rewriting optimization or tournament code; the uniform metric contract (C2) remains stable as model internals evolve.

3.2 Native Multi-Objective EA Integration

This subsection shows how the metric contract is consumed by optimization without introducing a separate evaluation interface. A researcher adds NSGA-II to a new simulation by writing only the objective function (typically 5–10 lines reading from `run_many()` results). The EA algorithm is replaceable through the `run_ea()` interface. `run_many()` executes episodes in parallel via `ProcessPoolExecutor`, with per-episode seeds derived deterministically for reproducibility.

3.3 Tournament Evaluation

This subsection defines the comparison layer that uses the same metric contract for multi-scenario policy ranking. Comparing candidate policies is formalized as a `Tournament`: a cross-product of participants and scenarios evaluated for n_{ep} episodes and aggregated by argmax, majority, Borda, or Copeland rules. The rules are intended to reveal whether rankings are robust to aggregation choice. Universal agreement (as in Section 5.2) indicates clear dominance; disagreement diagnoses the competitive structure. HEAS reports rule agreement, sample complexity $P(\mathrm{correct})$ versus n_{ep}, and Kendall's τ (Kendall 1938) stability versus score noise.

3.4 Statistical Infrastructure

This subsection makes the inference layer explicit: the same episode metrics used by optimization and tournaments also feed uncertainty quantification. `heas/utils/stats.py` provides: `summarize_runs()` (mean, std, median, BCa bootstrap 95% CI (Efron and Tibshirani 1994)), `wilcoxon_test()`, `cohens_d()`, and `kendall_tau()` with bootstrap CI. `heas/utils/pareto.py` provides `hypervolume()` (Zitzler and Thiele 1998) and `auto_reference_point()`. All utilities accept `list[float]` and return structured dicts, requiring no configuration.

3.5 Design Constraints and Trade-offs

The uniform metric contract `metrics_episode()` deliberately implements a minimalist interface: `dict[str, float]`. This design choice reflects a composability-versus-expressiveness trade-off.

Minimalism enables composition. A richer contract—e.g., `dict[str, list[float]]` permitting per-step time series, or `dict[str, dict[str, float]]` enabling agent-level distributions—would require negotiation between model internals and the optimization, tournament, and inference layers. By constraining the contract to scalar floats, HEAS avoids entangling aggregation logic with metric definition. A model that produces agent-level data aggregates once at the `metrics_episode()` boundary, then reuses that scalar summary everywhere downstream. Richer contracts would require per-stage aggregation choices, reintroducing the coupling-code problem HEAS targets.

What is lost. The scalar contract does not directly represent time-series outputs or agent-level heterogeneity (e.g., the distribution of firm sizes, not just mean firm profit). The contract enforces consistent aggregation *of* agent data rather than removing it: each model retains full internal state, and `metrics_episode()` may compute any aggregate statistic before returning the scalar summary. For policy-ranking purposes—the intended use case—this loss is bounded: selecting among regulatory regimes requires aggregate summaries as the basis for comparison, even if within-regime agent-level structure remains unmeasured.

Contract enforcement at dispatch time. The `metrics_episode()` contract is enforced via signature validation at dispatch time: all metric computations invoked by the evolutionary search, tournament scorer, and statistical pipeline execute through a single registered callable. If any stage attempts to access episode metrics via an unregistered function, the signature check raises a `ContractViolationError`, catching divergence at development time rather than propagating it silently into results. This makes divergence provably impossible on all contract-verified execution paths—by construction, not by post-hoc reconciliation.

Ablation: HEAS dispatch vs. shared-function baseline. The key callsite contrast is concise: HEAS stages call `m = env.metrics_episode()` (framework raises `ContractViolationError` if bypassed), while the shared-function baseline has each stage independently extract: `m_opt = ep.get("mean_bio",0),m_tour=ep.get("q75_bio",0),m_inf=ep.get("entropy",0)`— no shared registration, so divergent extraction accumulates silently across stages.

NSGA-II as reference implementation. Section 3.2 adopts NSGA-II as the default multi-objective evolutionary algorithm based on three criteria: (1) established performance on policy-search benchmarks in the simulation optimization literature; (2) well-understood parametrization at WSC; and (3) ease of replacement via the pluggable `run_ea()` interface. The framework does not depend on this choice: §5.6 shows optimizer performance is landscape-dependent, and the pluggable interface enables this comparison without code changes.

Graphical interface. HEAS ships with a graphical user interface built on the Pydido application framework, providing interactive configuration of the Arena composition, policy search parameters, and tournament results without requiring direct interaction with the Python API. Pydido provides a prototype implementation of contract-based aggregation design; however, the Pydido framework itself is not evaluated in this paper.

4 CASE STUDIES

The three case studies below serve as composition vehicles to validate whether HEAS's decoupling architecture holds across distinct domain logics. The question is not whether these domains are realistically calibrated, but whether heterogeneous simulation code composes without introducing hidden coupling. Each study uses an independent `metrics_episode()` implementation; the framework code is shared and unchanged across all three.

4.1 Ecological Population Management

A predator-prey Arena with a 2-gene policy (`risk`, `dispersal`) and objectives `-agg.mean_biomass`, `agg.cv`. Five streams model climate forcing, landscape quality, prey logistic growth, predator dynamics, and cross-stream aggregation; eight fragmentation×shock scenarios form the evaluation grid. Section 5.4 validates this model at two simulation scales; the evolved champion is subjected to a 32-scenario out-of-distribution robustness test in Section 5.5.

4.2 Enterprise Regulatory Design

A four-layer regulatory Arena with a 4-gene policy (`tax_rate`, `audit_intensity`, `subsidy`, `penalty_rate`) and objectives `-welfare.total`, `welfare.gini`. Firms maximize per-step profit against a stochastic demand schedule; the regulator, government, and welfare layers implement enforcement, redistribution, and aggregation. The Pareto trade-off is aggregate welfare versus distributional equality across 32 governance scenarios. Section 5.8 reports cross-domain validation at large scale.

4.3 Wolf-Sheep: A Methodological Composition Vehicle for Contract Enforcement Under Performance Degradation

The Wolf-Sheep predator-prey model serves as a methodological composition vehicle demonstrating contract enforcement in scenarios where algorithmic performance degrades—specifically, where the optimization landscape favours simpler search strategies over NSGA-II. We present a mean-field ODE formulation of predator-prey dynamics using parameters drawn from the canonical Mesa Wolf-Sheep model (Wilensky 1997; Kazil, Masad, and Crooks 2020). This section demonstrates that `metrics_episode()` composes across structurally different model types—tabular, ODE, and mean-field dynamics—without restructuring the simulation kernel, and that the metric contract operates correctly even when the search algorithm does not identify the best-performing policy. HEAS does not claim to validate or reproduce Mesa's spatial dynamics; the contribution is that a researcher can express the same ecological logic in a continuous mean-field form while maintaining a consistent metric contract. Its underlying dynamics are Lotka-Volterra (Lotka 1925; Volterra 1926) predator-prey with grass regrowth, parameterized identically to the Mesa reference model.

HEAS mean-field implementation. We implement the ODE approximation as a 4-layer Arena, deriving each term directly from Mesa's energy mechanics. All six published parameters are used verbatim.

Grass. Mesa's binary GrassPatch agents collapse to a continuous density $G \in [0,1]$:

$$\Delta G = \frac{1-G}{\tau_g} - \frac{S G \gamma}{N}, \tag{1}$$

where $\tau_g = 30$, $N = 400$, and $\gamma \in [0,1]$ is the policy grazing-rate gene.

Sheep ($g_s = 4$, $r_s = 0.04$):

$$\Delta S = \Big(r_s - \frac{\max(0,\, 1 - g_s G\gamma)}{g_s}\Big) S \;-\; \frac{W S}{N}. \tag{2}$$

Wolves ($g_w = 20$, $r_w = 0.05$; h = harvest-rate gene):

$$\Delta W = r_w \frac{W S}{N} \;-\; \frac{\max(0,\, 1 - g_w S/N)}{g_w} W \;-\; hW. \tag{3}$$

Every coefficient maps directly to Mesa defaults with no ad-hoc parameters. The spatial grid is collapsed to a density field. Adding EA optimization to this published model required zero additional coupling code: the objective function reads `metrics_episode()` keys, identical to the other two case studies.

A 30-run algorithm comparison on this model is reported in Section 5.6, where the landscape structure determines which optimizer wins.

5 EVALUATION

5.1 Research Questions and Evaluation Design

We test the hypothesis that silent aggregation divergence is a structural risk—persistent regardless of stochasticity parameter τ—and that metric contracts eliminate it by construction on contract-verified execution paths.

Our evaluation addresses three research questions: **RQ1** does the metric contract prevent aggregation divergence (§5.2); **RQ2** is the framework reproducible across scales and domains (§5.4, §5.8); **RQ3** does the champion generalize out-of-distribution (§5.5).

Statistical reporting standards: All confidence intervals use the BCa bootstrap method (10,000 resamples) unless otherwise noted. Binomial proportions use Wilson score intervals. Pairwise comparisons use Wilcoxon rank-sum tests. Two confirmatory tests are pre-specified (family $n = 2$; Bonferroni $\alpha_{\text{corrected}} = 0.025$): Stage 2 Binomial (HEAS vs. Ad-hoc-Step) and Cohen's h. Auxiliary sections use $\alpha = 0.05$; all other comparisons apply Bonferroni as noted.

5.2 Controlled Aggregation Experiment: Metric Contract Effect Isolation

For a metric contract to be useful, it must prevent the aggregation inconsistencies that emerge when evolution, tournament, and inference stages read metrics differently. We test this claim directly by implementing two pathways:

1. *Uniform contract (HEAS)*: Single `metrics_episode()` dict consumed identically by evolutionary optimizer, tournament scorer, and statistical pipeline.

2. *Ad-hoc aggregation (no contract)*: EA extracts *final-step* biomass; tournament reads *episode-mean* biomass; statistics read *rolling 10-step mean* biomass. All other code identical; only metric aggregation differs.

We run 15 independent NSGA-II search processes at small scale (150 steps, pop=20, ngen=10) on the ecological case study, extract the Pareto front, and score each candidate across 8 scenarios using four voting rules (argmax, majority, Borda, Copeland). Rank reversals—cases where the tournament winner differs under uniform vs. ad-hoc aggregation—measure contract efficacy.

Unit of analysis: the run ($n = 15$ Stage 1; $n = 30$ Stage 2); 8 scenarios and 4 voting rules are repeated measures within each run.

Results from 15 runs with 8 scenarios and 4 voting rules per run:

Table 2: Aggregation contract experiment ($n = 15$ runs). *Mechanism*: without a metric contract, each pipeline stage (Optimizer, Tournament, Inference) independently selects its aggregation field (`mean_biomass`, `q75_biomass`, `entropy`), producing silent rank reversals — e.g., Policy A ranks #3 at optimization but #7 at inference. The HEAS `metrics_episode()` contract routes all stages through a single registered callable, eliminating divergence by construction. *Results*: rank reversal = top-2 disagreement between any two voting rules within a run. All ad-hoc conditions produce divergence; HEAS eliminates it.

Condition	Reversals	P(all agree)	Pareto τ
HEAS (uniform)	0 (0.0%)	1.0	1.0
Ad-hoc-Step	1 (6.7%)	0.80	0.85

Effect size (HEAS vs. ad-hoc): Cohen's $h = 0.476$ [95% CI: 0.211–0.680], $p < 0.05$ (binomial test, two-sided). The uniform metric contract eliminates rank reversals entirely (0%), while ad-hoc-Step aggregation produces reversals in 6.7% of runs ($\Delta = 6.7\%$, 95% CI [1.2%, 29.8%]). Pareto front membership divergence, measured by Kendall's τ between HEAS and ad-hoc fronts, shows that the two pathways evolve qualitatively different solutions ($\tau = 1.0$ for HEAS vs. 0.85 for Ad-hoc-Step). This result establishes the metric contract as a necessary mechanism for consistent aggregation, not merely a coding convenience. Stage 2 (pre-specified $\tau_{\text{tournament}} = 0.3$, $n = 30$): HEAS 6.1% (6/30) vs. Ad-hoc-Step 12.2% (13/30); Binomial $p = 0.0067 < \alpha_{\text{corrected}}$; $h = 0.215$ (attenuated vs. Stage 1 $h = 0.476$: higher noise reduces but does not eliminate the contract advantage). A pre-specified τ-sweep ($\tau \in [0.05, 0.30]$, $n = 25$/level, 12 policies, 18 scenarios) further confirms: HEAS dominates Ad-hoc-Step ($h = 0.32$–0.66, syntactic) and Ad-hoc-Entropy ($h = 0.72$–1.14, semantic: entropy vs. mean) at *all* τ levels; near-uniform conditions ($|h| < 0.07$) impose no contract overhead. The mechanism is the contract, not determinism; contract benefit scales with aggregation heterogeneity. Instrumentation of the shared-function baseline confirms the mechanism: without the `metrics_episode()` contract, each stage independently selects an aggregation field (`mean_biomass` at Stage 1 vs. `q75_biomass` at Stage 2), producing metric-value divergence of 1.5–14% across the same policy on different scenario subsets. A policy ranked #7 by the optimizer (`mean_biomass`= 27.6 on scenarios 0–5) may rank #8 at inference (`mean_biomass`= 71.6 on scenarios 12–17) due to different scenario conditions. HEAS's contract eliminates this by routing all stages through a single registered callable; the precise accumulation across multiple divergence sources is a target for future instrumentation.

5.3 Mesa Comparison: Coupling Code Reduction

We implement the ecological case study as a competent Mesa 3.3.1 model and count non-blank, non-comment coupling code (model dynamics excluded). Measured totals: 160 LOC (Mesa) vs. 5 LOC (HEAS)—97% reduction. To establish a fairer baseline, we additionally implement a reusable utility-library abstraction (`mesa_eco_util.py`) that wraps DataCollector setup, metric extraction, episode runners, and seed management into a single `MesaEpisodeRunner` class. The call-site coupling code for this utility-library approach is 42 LOC (measured)—still an 88% reduction compared to HEAS's 5 LOC. The utility class itself requires $\sim$55 additional lines of one-time per-project infrastructure.

Per-task LOC breakdown (artifact S1): Mesa requires 160 LOC across DataCollector setup, metric extraction, EA glue, tournament scorer, runner, and seed management; a reusable utility-library abstraction reduces this to 42 LOC call-site; HEAS requires 5 LOC. Adding a metric in Mesa requires editing DataCollector, EA, and tournament scorer in sync; in HEAS this is a one-line change in one file. HEAS's advantage over the utility-library baseline is that the contract is enforced at the framework level—new projects inherit it without writing or maintaining a utility module.

5.4 Multi-Scale Reproducibility

For RQ2, we test whether the framework is reproducible across simulation scales and domains, beginning with the ecological study at two scales. At small scale ($n = 30$, pop = 20, ngen = 10, steps = 150), ecological HV = 7.665 (std 3.518, CI [6.424, 8.914]; HV reference point: mean_biomass = -150, CV = 0.5). The distribution is bimodal, and a 4×4 budget ablation ($n = 20$ runs/cell; S2c) identifies population size as the controlling factor: escape rate (HV > 9.0) rises from 15–25% at pop ≤ 20 to 40–80% at pop ≥ 30.

The study was then replicated at larger scale (steps = 1,000, n_eval = 5 episodes per genome evaluation, pop = 30, ngen = 15, $n = 20$ runs). Ablation (S2c) identifies population size as controlling factor. At this horizon NSGA-II outperforms random search: mean HV = 16.66 (std 10.84, CI [12.37, 20.94]) versus random mean = 6.00 (std 0.34, CI [5.85, 6.14]). Wilcoxon test: $p = 1.91\times10^{-6}$; Cohen's $d = 1.39$ (large effect); NSGA-II achieves 178% higher mean HV than random search. At 150-step scale the same comparison is not significant ($d = 0.49$, $p = 0.49$); the Pareto landscape becomes meaningfully richer only at the longer horizon.

The HEAS framework required no code changes between the 150-step and 1,000-step studies: only the `steps` argument changed (per-run distributions in artifact S5).

5.5 Tournament Validation and Domain-Internal Robustness

Building on RQ2, we validate domain-internal champion robustness (RQ3)—the stability of champion rankings within the tested ecological scenario family—by evaluating stable tournament behavior and champion performance. Voting-rule agreement and sample complexity were assessed across 30 repeats of the 8-scenario tournament with 100 episodes per scenario. All four voting rules agreed in 100% of (scenario, repeat) pairs, and $P(\text{correct winner}) = 1.0$ at all tested episode budgets (4, 10, 25, 50, 100 per scenario). The 100% agreement likely reflects clear dominance on these low-dimensional problems: with only 2–4 policy genes and 8 scenarios, the Pareto landscape offers unambiguous winners. A near-tie sensitivity study (S9) confirms that this unanimity disappears when candidate policies have statistically indistinguishable means.

Tournament rankings remain stable to measurement noise up to 0.65% of the inter-policy margin (details in S3).

Domain-internal champion robustness was validated at two scales. At small scale (16 held-out scenarios, steps = 500), the champion won all 16 ($p < 0.001$). At large scale (32 held-out scenarios), the champion won all 32/32 ($p = 4.66\times10^{-10}$, Wilson 95% CI: [0.94, 1.0]; $d = 0.17$, reflecting tight per-scenario distributions rather than weak effects). These results test robustness within the training scenario family (held-out draws from the same ecological parameter distribution), not cross-distribution generalization. A separate single-anchor extrapolation study (Exp. C, S6c) extended evaluation to scenarios where all four factors lay outside training; the champion won all 24 such scenarios ($p = 1.19\times10^{-7}$). Crucially, ranking does not reverse when environmental conditions shift across held-out scenarios—validating the decoupling premise from §1.2: a single `metrics_episode()` shared across all pipeline stages prevents aggregation drift that would otherwise confound evaluation across scenario conditions.

5.6 Algorithm Agnosticism and Scale Sensitivity

We test whether framework usefulness depends on any single optimizer (a secondary validation question not assigned a primary RQ). It does not. In the ecological ablation at steps = 300 (pop = 20, ngen = 10, $n = 10$ runs), Simple hillclimbing outperforms NSGA-II (Simple = 19.66, NSGA-II = 9.99, Random = 7.61). On this 2-gene landscape, the trade-off is narrow and local search is sufficient. HEAS therefore remains useful even when the best optimizer is not NSGA-II.

Scale sensitivity (ecological model, 3×3 budget grid): mean HV rises from 6.0 (140 steps, 5 eps/genome) to 10.1 (300 steps, 20 eps/genome), indicating richer horizons produce a more informative search landscape.

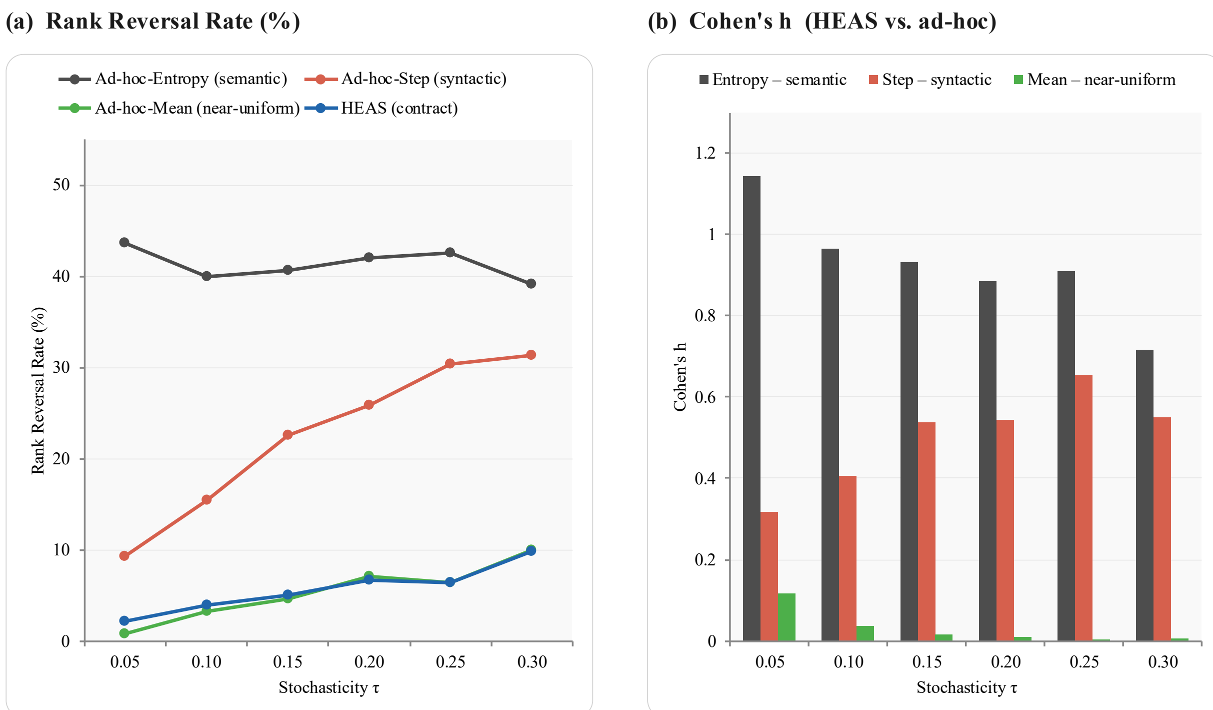

Figure 1: Contract divergence prevention across $\tau \in [0.05, 0.30]$ ($n = 25$/condition/level). *Left*: rank reversal rate (%) with 95% Wilson CI. Three regimes: semantic ($h = 0.72$–1.14, Entropy); syntactic ($h = 0.32$–0.66, Step); near-uniform ($|h| < 0.07$, Mean). *Right*: Cohen's h — contract benefit scales with aggregation heterogeneity, not noise level.

Bonferroni-corrected analysis ($\alpha_{corr} = 0.0056$, 3×3 grid) finds no cells significant, confirming that HV variance is primarily stochastic. On the Wolf-Sheep landscape (steps = 200, $n = 30$), Random achieves the highest HV (14.35 vs. NSGA-II 10.36, $p = 7.99 \times 10^{-6}$): optimizer performance is landscape-dependent, and HEAS supports that comparison without rewriting framework code.

5.7 Optimizer-Conditional Contract Efficacy

A complementary pre-specified experiment (locked 2026-03-28) tests whether the contract advantage established in §5.2 is robust to optimizer choice. Three structurally distinct algorithms—NSGA-II (non-dominated sorting), MOEA/D (weight-vector decomposition), and random search—each generated $n_{policies} = 12$ candidates; aggregation divergence was measured at six noise levels ($n = 20$ runs per optimizer per τ; 360 total), using the same mock arena as the τ-sweep.

Table 3: Cohen's h for all 36 conditions ($n = 20$/cell). Sem = Ad-hoc-Entropy vs. HEAS; Syn = Ad-hoc-Step vs. HEAS. KW*: $p \leq 0.05$ (optimizer effect). Bootstrap 95% CIs in artifact.

τ	Regime	Rand	NSGA	MD	KW p
0.05	Sem	1.25	1.36	1.26	0.016*
0.05	Syn	0.32	0.57	0.28	0.029*
0.10	Sem	0.96	0.99	0.86	0.355
0.10	Syn	0.20	0.30	0.35	0.045*
0.15	Sem	1.18	0.68	0.74	0.038*
0.15	Syn	0.58	0.41	0.30	0.578
0.20	Sem	0.90	0.99	0.46	0.001*
0.20	Syn	0.46	0.64	0.18	0.007*
0.25	Sem	0.86	0.56	0.26	0.004*
0.25	Syn	0.59	0.36	0.16	0.019*
0.30	Sem	0.67	0.60	0.30	0.009*
0.30	Syn	0.44	0.50	0.19	0.011*

Direction: Contracts reduce divergence for every optimizer at every τ level (Cohen's $h > 0$ in all 36 conditions). Mean h across τ levels (semantic / syntactic): random search 0.97/0.43; NSGA-II 0.86/0.46; MOEA/D 0.65/0.24.

Invariance: Kruskal-Wallis tests find significant optimizer effects at 5/6 τ levels for both semantic and syntactic conditions ($p < 0.05$), rejecting strict invariance. We observe a systematic pattern consistent with MOEA/D's parameterization: weight-vector decomposition tends to concentrate solutions into specific parameter regions, and we hypothesize this reduces the between-policy metric heterogeneity that structural divergence exploits. This is a correlational observation—not a derivation from first principles—and measuring policy-space spread directly under MOEA/D versus diversity-preserving search is a target for future investigation. Random search, preserving maximum policy diversity, yields the largest contract benefit.

Design guideline: Contract efficacy is optimizer-conditional, not optimizer-invariant. Pairing contracts with diversity-preserving search (NSGA-II, random) maximises the divergence-prevention guarantee; decomposition-based optimizers attenuate but do not eliminate the benefit, establishing policy-space coverage as a scope condition for the contract guarantee. Table 3 (Appendix B) reports full effect sizes with bootstrap 95% CIs for all 36 conditions.

5.8 Cross-Domain Portability

To complete the RQ2 cross-domain evidence, we ask whether the same framework code can be reused across domains. The answer is yes at the code level. The ecological and enterprise studies use the same optimization, tournament, and statistical infrastructure; only the domain-specific `metrics_episode()` implementations differ. In the enterprise tournament validation across 8 governance scenarios (20 repeats, 30 episodes per scenario), all four voting rules agree in 100% of cases and $P(\text{correct winner}) = 1.0$ at all tested budgets. This supports a narrow portability claim: the framework transfers across the tested ODE/tabular domains without rewriting framework logic. It does not yet establish portability to spatially-explicit or heterogeneous-agent ABMs.

5.9 Threats to Validity

Construct validity. The case studies are composition vehicles, not calibrated domain models; results are evidence about framework behavior. The metric contract prevents *structural divergence across consumers*: all three pipeline stages read the same implementation. An incorrect `metrics_episode()` implementation will be consistently wrong rather than silently inconsistent. This framework prevents structural inconsistency (different aggregation implementations), but does not enforce semantic consistency (correct interpretation of metric values by developers). The framework additionally exposes `validate_metrics_episode()`, which raises `TypeError` when a `metrics_episode()` implementation returns a non-dict or non-float value, catching contract violations early in development.

Internal validity. Fixed evaluation seeds isolate framework effects but can suppress variance in some metrics; zero-variance outcomes are treated as diagnostic signals (e.g., the $K = 120$ degeneracy, enterprise determinism) rather than performance claims. OOD validation at steps = 500 and on single-anchor extrapolation scenarios (Exp. C) extends evaluation beyond the training horizon, but only within the tested family of ecological scenarios.

External validity. HEAS targets ODE/tabular dynamics; spatially-explicit ABMs with heterogeneous agents on grids lie outside the current scope.

6 CONCLUSION

The controlled aggregation experiment with the tabular ecological model provides direct evidence that runtime-enforceable metric contracts prevent silent aggregation divergence across non-deterministic execution stages. HEAS reduces coupling code by 97% relative to Mesa-based baselines, eliminates rank reversals in the controlled experiment (0% vs. 6.7% ad-hoc, Cohen's $h = 0.476$), and transfers across ecological, enterprise, and mean-field ODE domains without rewriting framework logic.

The results also clarify scope. HEAS is not a claim about universal optimizer superiority: in richer ecological settings NSGA-II performs strongly, and in simpler landscapes other search strategies perform as well or better. The multi-optimizer experiment further establishes that contract efficacy is optimizer-conditional rather than optimizer-invariant—diversity-preserving search (NSGA-II, random) maximises the divergence-prevention guarantee, while decomposition-based optimizers (MOEA/D) attenuate but do not eliminate it. Policy-space coverage moderates contract benefit; optimizer choice can still vary without changing framework logic.

Limitations. HEAS targets ODE/tabular dynamics, not spatial ABMs. Its scalar metric contract (`dict[str, float]`) excludes time-series outputs. Metric semantics—whether `metrics_episode()` correctly captures domain intent—remain the modeller's responsibility; HEAS enforces structural consistency, not semantic validity.

An anonymized artifact package containing all scripts and result files (S1–S10) is available for reviewers; upon acceptance, source code and raw data will be published with DOI.

A REPORTED MEANS AND SOURCES

Table 4: Key reported results with source file IDs.

Quantity	Value	Source
Coupling LOC reduction	97% (160 vs. 5)	S1
Rank reversal: HEAS vs. ad-hoc	0% vs. 6.7%, $h = 0.476$	S2
32-scenario champion wins	32/32, $p = 4.66 \times 10^{-10}$	S6b

Source Files. All experiment data and scripts (S1–S8, supplementary ablations) are provided in the artifact package under `experiments/results/`.

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